

Client Summary

E-Commerce RFM Customer Segmentation — Engagement Overview

Client Context: Multi-category e-commerce marketplace (Olist, Brazil) **Engagement Type:** Customer Behavioral Segmentation & CRM Analytics **Analyst:** Mohammad Ammar — Principal Data Analytics Consultant

1. The Business Question

Your marketing and CRM teams are treating a diverse customer base with uniform, blast-style campaigns. This engagement answers a simple question with rigorous data: **which customers matter most, which are about to leave, and what should you do about each group?**

Details of the underlying business problem are documented in docs/03_Business_Problem.md.

2. What We Analyzed

- **94,989 unique customers**, identified consistently across multiple orders via `customer_unique_id`
 - **R\$ 15,737,667.52** in total gross revenue (product price + freight)
 - Transaction history from September 2016 through October 2018
 - Five relational data tables joined at the customer level (see docs/09_Data_Model.md)
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3. What We Found

Your customer base splits into five clear behavioral groups:

Segment	Who They Are	Size	Revenue Contribution
Champions	Recent, frequent, high-spend buyers	13,354 customers	R\$ 4,089,088.19 (25.98%)
Loyal	Consistent repeat buyers	41,740 customers	R\$ 5,054,591.30 (32.12%)
At Risk	Previously high-value, now going quiet	1,900 customers	R\$ 427,933.15 (2.72%)
Lost	Inactive, single-purchase buyers	12,593 customers	R\$ 2,073,734.79 (13.18%)
Others	Moderate activity, unclassified pattern	25,402 customers	R\$ 4,092,320.09 (26.00%)

The headline finding: just under 58% of your customers (Loyal + Champions) generate 58.10% of your revenue — and your Champions spend, on average, 2.53x more than your Loyal customers (R\$ 306.21 vs. R\$ 121.10 ARPU). Treating these two groups the same way in a marketing calendar leaves money on the table in both directions.

Full findings: docs/12_Business_Insights.md.

4. Where the Money Is at Risk

Two groups require immediate attention:

1. **1,900 At-Risk customers** represent R\$ 427,933.15 in revenue that is actively slipping away.

These are not low-value customers — their average spend (R\$ 225.23) is the second-highest of any segment.

2. **12,593 Lost customers** represent R\$ 2,073,734.79 in historical spend that has already gone cold. Reactivating even 5% of this group recovers over R\$ 103,000 in gross sales — at effectively zero acquisition cost, since these are existing customer relationships.

5. What We Recommend

Segment	Recommended Action
Champions	VIP loyalty tier, early access, dedicated support — protect the highest-margin revenue base
Loyal	Cross-sell and upsell campaigns to lift average spend toward Champion-level behavior
At Risk	Automated 14-day inactivity triggers with time-bound offers before customers go fully dormant
Lost	Low-cost automated win-back email sequences highlighting previously purchased categories
Others	Onboarding and second-purchase incentives to move customers into the Loyal tier

Detailed campaign design and execution phasing: docs/13_Business_Recommendations.md.

6. How This Was Built

- **SQL pipeline:** sql/01_rfm_metrics.sql, sql/02_customer_segmentation.sql — window-function scoring (NTILE(5)) with custom frequency binning to handle single-order skew
 - **Python pipeline:** python/rfm_analysis.py — reproducible, modular processing
 - **Dashboard:** Interactive Tableau dashboard for ongoing self-service exploration
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7. Honest Limitations

This analysis is transactional, not predictive — it tells you what customers have *done*, not what they will do next. It does not include demographic data, web behavior, or forecasting models. See docs/14_Limitations.md for the full constraints list, including why recency is calculated against a fixed historical snapshot rather than a live date.

8. Next Steps

1. Review segment-level campaign triggers with CRM and performance marketing leads
2. Prioritize At-Risk intervention given its urgency and ARPU profile
3. Establish this RFM model as a recurring (not one-time) analytical process in the data warehouse

Questions on any figure in this summary can be traced directly to the SQL logic in sql/ or the methodology in docs/10_Methodology.md.